viewed as a map U(t) which takes as input $S(t_0)$ and produces $S(t + t_0)$, that is: $S(t + t_0) = U(t)S(t_0)$. We say that the laws giving rise to U(t) are time-reversal symmetric if there is a map T satisfying $U(-t) = T^{-1}U(t)T$. In English, this equation says that by a suitable manipulation of the state of the physical system at one moment (accomplished by **T**), evolution by an amount *t* forward in time according to the laws of the theory (accomplished by U(t)) is equivalent to having evolved the system t units of time backward in time (denoted by U(−t)). For instance, if we specify the state of a system of particles at one moment by their positions and velocities, then **T** would keep all particle positions fixed and reverse all velocities. Evolving such a configuration of particles forward in time by an amount *t* is equivalent to having evolved the original configuration of particles backward in time by an amount t. (The factor of T^{-1} undoes the velocity reversal so that, at the end, not only are the particle positions what they would have been *t* units of time previously, but so are their velocities.)

For certain sets of laws, the **T** operation is more complicated than it is for Newtonian mechanics. For example, if we study the motion of charged particles in the presence of an electromagnetic field, the reversal of particle velocities would be inadequate for the equations to yield an evolution in which the particles retrace their steps. Instead, the direction of the magnetic field must also be reversed. (This is required so that the $\boldsymbol{v} \times B$ term in the Lorentz force law equation remains unchanged.) Thus, in this case, the **T** operation encompasses both of these transformations. The fact that we have to do more than just reverse all particle velocities has no impact on any of the discussion that follows in the text. All that matters is that particle motion in one direction is just as consistent with the physical laws as particle motion in the reverse direction. That we have to reverse any magnetic fields that happen to be present to accomplish this is of no particular relevance.

Where things get more subtle is the weak nuclear interactions. The weak interactions are described by a particular quantum field theory (discussed briefly in Chapter 9), and a general theorem shows that quantum field theories (so long as they are local, unitary, and Lorentz invariant—which are the ones of interest) are always symmetric under the *com*bined operations of charge conjugation C (which replaces particles by their antiparticles), parity P (which inverts positions through the origin), and a bare-bones time-reversal operation *T* (which replaces *t* by −*t*). So, we could define a **T** operation to be the product *CPT,* but if **T** invariance absolutely requires the *CP* operation to be included, **T** would no longer be simply interpreted as particles retracing their steps (since, for example, particle identities would be changed by such **T—**particles would be replaced by their antiparticles—and hence it would not be the original particles retracing their steps). As it turns out, there are some exotic experimental situations in which we are forced into this corner. There are certain particle species (K-mesons, B-mesons) whose repertoire of behaviors is *CPT* invariant but is not invariant under *T* alone. This was established indirectly in 1964 by James Cronin, Val Fitch, and their collaborators (for which Cronin and Fitch received the 1980 Nobel Prize) by showing that the K-mesons violated CP symmetry (ensuring that they must violate *T* symmetry in order *not* to violate *CPT*). More recently, *T* symmetry violation has been directly established by the CPLEAR experiment at CERN and the KTEV experiment at Fermilab. Roughly speaking, these experiments show that if you were presented with a film of the recorded processes involving these meson particles, you'd be able to determine whether the film was being projected in the correct forward time direction, or in reverse. In other words, these particular particles can distinguish between past and future. What remains unclear, though, is whether this has any relevance for the arrow of time we experience in everyday contexts. After all, these are exotic particles that can be produced for fleeting moments in high-energy collisions, but they are not a constituent of familiar material objects. To many physicists, including me, it seems unlikely that the time nonreversal invariance evidenced by these particles plays a role in